People's Democratic Republic of Algeria

Ministry of National Defence

Military Polytechnic School



Department: Computer Science

Specialization: Networks and Distributed Systems

Data Mining Mini-Project Report

Intelligent Firewall Traffic Analyzer: Prediction of Firewall Actions Using Data Mining

---

Prepared by:

RSD-022

RSD-009

RSD-020

RSD-002

Academic Year:
2026/2027

Acknowledgements

We would like to express our deepest gratitude and sincere appreciation to our professor, Dr. Adil Hosni, for his dedication, continuous efforts, and invaluable guidance throughout this course. We are equally grateful for his kindness, generosity, and genuine consideration towards his students, which made our learning experience all the more meaningful and rewarding.

It has truly been an honor and a privilege to have been among his students. We deeply appreciate not only the knowledge he has shared with us but also the exemplary character he has demonstrated throughout our time learning under his guidance.

Table of Contents

Acknowledgements **ii**

List of Figures **v**

List of Tables **vi**

List of Acronyms **viii**

General Introduction **1**

I Context and Problem Statement **2**

I.1 Introduction 2

I.2 Firewalls and Traffic Logs 2

I.3 Problem Statement and Objectives 2

I.4 Project Scope and Constraints 2

I.5 Related Work 2

I.6 Conclusion 2

II Theoretical Foundations **3**

II.1 Introduction 3

II.2 Data Mining Process 3

II.3 Supervised Classification and Selected Models 3

II.3.1 Baseline Classifier (Zero-R) 3

II.3.2 K-Nearest Neighbors 3

II.3.3 Decision Tree and Random Forest 3
II.3.4 Naive Bayes 3
II.3.5 Support Vector Machines 4
II.4 Model Evaluation 4
II.5 Cross-Validation and Hyperparameter Tuning 4
II.6 Statistical Tests 4
II.7 Class Imbalance, Duplicates, and Data Leakage 4
II.8 Conclusion 4

III Dataset and Exploratory Data Analysis **5**
III.1 Introduction 5
III.2 Dataset Source and Description 5
III.3 Data Quality Audit 5
III.4 Univariate Analysis 5
III.5 Bivariate Analysis and Correlations 5
III.6 Statistical Tests 5
III.7 Summary and Identified Risks 6
III.8 Conclusion 6

IV Experimental Methodology **7**
IV.1 Introduction 7
IV.2 Task Selection and Feature Availability 7
IV.3 Preprocessing 7
IV.4 Data Splitting 7
IV.5 Experimental Protocol 7
IV.6 Conclusion 7

V Experiments and Results **8**
V.1 Introduction 8

V.2 Experimental Environment 8
V.3 Cross-Validation Results 8
V.4 Test Set Results 8
V.5 Rare Class and Data Leakage Features 8
V.6 Discussion 8
V.7 Conclusion 9

VI Interactive Application **10**
VI.1 Introduction 10
VI.2 Architecture 10
VI.3 Application Pages 10
VI.4 Prediction Form and Input Validation 10
VI.5 Work Organization and Git Traceability 10
VI.6 Conclusion 10

General Conclusion and Future Work **11**

References **12**

Appendix A: Hyperparameter Grids **13**

Appendix B: Detailed Results **14**

Appendix C: Reproducing the Results **15**

List of Figures

List of Tables

List of Acronyms

General Introduction

[TODO : Context (1 paragraph): the role of a firewall and its logs, and the value of analyzing them using data mining]

[TODO : Problem Statement (1 paragraph): the research question and challenges (class imbalance, duplicates, data leakage)]

[TODO : Approach and Contributions (1 paragraph)]

To facilitate reading, this report is organized into six chapters:

- **Chapter I** presents the context and problem statement.
- **Chapter II** reviews the theoretical foundations.
- **Chapter III** describes the dataset and its exploratory analysis.
- **Chapter IV** details the experimental methodology.
- **Chapter V** presents and discusses the results.
- **Chapter VI** describes the interactive application.

CHAPTER I

CONTEXT AND PROBLEM STATEMENT

I.1 Introduction

[TODO : Chapter introduction]

I.2 Firewalls and Traffic Logs

[TODO : Definitions and the role of each concept in the project]

I.3 Problem Statement and Objectives

[TODO : Formal problem statement and objectives]

I.4 Project Scope and Constraints

[TODO : Selected constraints and project scope]

I.5 Related Work

[TODO : Related work using `\cite`]

I.6 Conclusion

[TODO : Chapter conclusion and transition]

CHAPTER II

THEORETICAL FOUNDATIONS

II.1 Introduction

[TODO : Chapter introduction]

II.2 Data Mining Process

[TODO : Process overview and its connection to the following chapters]

II.3 Supervised Classification and Selected Models

II.3.1 Baseline Classifier (Zero-R)

[TODO : Principle and role as a lower-bound baseline]

II.3.2 K-Nearest Neighbors

[TODO : Distance, choice of K, standardization, weighting]

II.3.3 Decision Tree and Random Forest

[TODO : Gini index, entropy, pruning, bagging, feature importance]

II.3.4 Naive Bayes

[TODO : Bayes' theorem, independence assumption, smoothing]

II.3.5 Support Vector Machines

[TODO : Margin, kernels, parameters C and gamma]

II.4 Model Evaluation

[TODO : Definitions with formulas (`equation`) and justified metric selection]

II.5 Cross-Validation and Hyperparameter Tuning

[TODO : Principles and justification of the evaluation protocol]

II.6 Statistical Tests

[TODO : Three tests: definition, explanation, and role in the project]

II.7 Class Imbalance, Duplicates, and Data Leakage

[TODO : Concepts and risks, with references]

II.8 Conclusion

[TODO : Chapter conclusion and transition to Chapter III]

CHAPTER III

DATASET AND EXPLORATORY DATA ANALYSIS

III.1 Introduction

[TODO : Chapter introduction]

III.2 Dataset Source and Description

[TODO : Table describing the 12 columns (based on Notebook 01)]

III.3 Data Quality Audit

[TODO : Audit figures and their interpretation]

III.4 Univariate Analysis

[TODO : Figures and interpretation]

III.5 Bivariate Analysis and Correlations

[TODO : Correlations, redundancy, and policy signatures]

III.6 Statistical Tests

[TODO : Results of the two tests and justification for their selection]

III.7 Summary and Identified Risks

[TODO : Brief summary in prose]

III.8 Conclusion

[TODO : Chapter conclusion and transition to Chapter IV]

CHAPTER IV

EXPERIMENTAL METHODOLOGY

IV.1 Introduction

[TODO : Chapter introduction]

IV.2 Task Selection and Feature Availability

[TODO : Selected task and feature availability table]

IV.3 Preprocessing

[TODO : Preprocessing pipeline with a diagram (draw.io) and justification for each step]

IV.4 Data Splitting

[TODO : Principle, verified counts, and data splitting diagram]

IV.5 Experimental Protocol

[TODO : Complete experimental protocol and hyperparameter grid table]

IV.6 Conclusion

[TODO : Chapter conclusion and transition to Chapter V]

CHAPTER V

EXPERIMENTS AND RESULTS

V.1 Introduction

[TODO : Chapter introduction]

V.2 Experimental Environment

[TODO : Software and hardware versions table]

V.3 Cross-Validation Results

[TODO : Results table and interpretation]

V.4 Test Set Results

[TODO : Results table, confusion matrices, and ROC curves]

V.5 Rare Class and Data Leakage Features

[TODO : Rare-class analysis and ablation study]

V.6 Discussion

[TODO : Discussion in prose]

V.7 Conclusion

[TODO : Chapter conclusion and transition to Chapter VI]

CHAPTER VI

INTERACTIVE APPLICATION

VI.1 Introduction

[TODO : Chapter introduction]

VI.2 Architecture

[TODO : Architecture diagram and component descriptions]

VI.3 Application Pages

[TODO : Annotated screenshots (with no personal names or file paths visible)]

VI.4 Prediction Form and Input Validation

[TODO : How the prediction form works and the limitations of its outputs]

VI.5 Work Organization and Git Traceability

[TODO : Table of responsibilities by student code]

VI.6 Conclusion

[TODO : Chapter conclusion]

General Conclusion and Future Work

Summary of the Work

[TODO : Summary of the work]

Limitations

[TODO : Honest discussion of the limitations]

Future Work

[TODO : Future directions]

References

Appendix A: Hyperparameter Grids

[TODO : Hyperparameter grids tested for each model]

Appendix B: Detailed Results

[TODO : Results for each fold and complete classification reports]

Appendix C: Reproducing the Results

[TODO : Installation and execution instructions (extracted from the README)]